

# OPERATING MANUAL

## Common to All Models

### Supplemental Explanation

Manual No.: HGENSA0062E

NC: MAZATROL SmoothAi  
MAZATROL SmoothX  
MAZATROL SmoothG  
MAZATROL SmoothEz  
MAZATROL SmoothC

Serial No.:

**Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.**

#### IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. If there is a difference between the contents of the manual and the actual state of the machine, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.
7. The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.
8. Unauthorized copying or reproduction of this manual is prohibited.

Issued by Manual Publication Section, YAMAZAKI MAZAK CORPORATION, Japan

01.2025

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## 1 SAFETY PRECAUTIONS

Although similar precautions are described in the Operating and Maintenance Manuals for each machine, read the precautions in this document carefully along with the precautions in the manuals for each machine, because they are especially related to operations in which the operator is directly involved.

### 1-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:



**DANGER**

: Failure to follow these instructions could result in loss of life.



**WARNING**

: Failure to observe these instructions could result in serious harm to a human life or body.



**CAUTION**

: Failure to observe these instructions could result in minor injuries or serious machine damage.

### 1-2 Precautions Related to Gripping Workpiece

Here are some safety precautions related to gripping a workpiece.

#### 1-2-1 General precautions and precautions related to attachments



**WARNING**

- Prior to start the operations of the machine, ensure the workpiece is securely clamped on to workpiece clamping device (for example; chuck, work rest, tailstock etc. for lathe, tables and fixtures for machining center).
- Never try to machine a workpiece heavier than the mass specified in the Operating Manual. Doing so may cause the workpiece to jump out. (For turning machines, the permissible mass of the workpiece is calculated by subtracting the mass of the chuck and the mass of the jaws from the maximum supporting mass of the machine specified in the specification sheet or the like.)
- Do not rotate the spindle with an attachment at a speed higher than the rated speed. If you are using a chuck or an attachment that is not recommended by us, make an inquiry to the manufacturer about the safety operation speed.
- Always use a lifting gear that matches the weight to be supported.

#### 1-2-2 Turning



**WARNING**

- When an unbalanced workpiece is cut by turning, it might fly out. To avoid this, balance the workpiece or set machining conditions carefully so as to reduce the spindle speed, etc.
- Do not rotate the workpiece holding device (e.g., the chuck or the circular pallet) at a speed higher than the speed specified in the "holding force - rotation speed diagram" provided by the manufacturer.
- In the parameter, do not enter a value that exceeds the allowable rotation speed of the chuck (or circular pallet).

**1-2-3 Long, heavy bar workpiece**

- If the center hole in the heavy bar workpiece is too small, the workpiece may jump out when a load is applied to it. So, perform a trial run to see that operation can be performed safely. (Lathe)
- Do not leave the power to the machine off for a long time when a workpiece is clamped in the spindle. (Lathe)
- Do not turn off the power when a workpiece is supported by the tailstock. If the power is left off for a long time, the supporting force of the tailstock may be reduced, thus causing the workpiece to fall. (Lathe)
- Do not rotate the spindle if the bar workpiece is protruding from the spindle, filler tube or bar feeder. Failure to observe this may result in a serious injury. Before starting the spindle, check to make sure that the bar workpiece is not protruding. (Lathe)
- Use a proper lifting gear or a crane to lift a heavy device or workpiece with a weight of more than 10 kg (22 lbs), but avoid lifting it by yourself.

**1-2-4 Machining center**

- Make sure that no device (vice or fixture) is protruding from the table.
- Be sure to clamp the pallet securely when setting or removing a workpiece.

**1-2-5 Lathe**

- After forming of the soft jaws, ensure that they fit the workpiece to clamp it securely in the chuck.
- Do not attach the chuck jaws in such a way that they will extend further out than the chuck outer diameter. Interference of the TOOL EYE and jaw may cause a failure of the machine.

**1-2-6 Turning spindle**

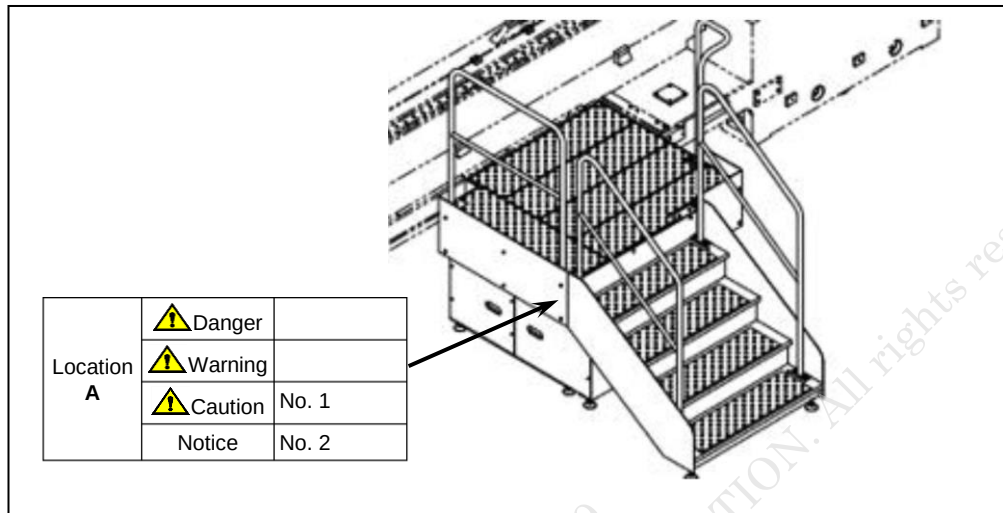
- If the spindle is directly rotated by hand at the time of setup, its rotation speed may sometimes exceed the safety speed. In that case, the alarm **2522 SPINDLE EXCEEDED SAFE SPEED** will be displayed for warning.  
To clear the alarm, press the reset key on the operating panel or tap the alarm clear button.

### 1-3 Safety Precautions Related to Stairs

Some models are equipped with stairs for accessing the operation and setup positions.

They have the following residual risk factors. So, if you are using one of them, always keep the residual risk factors in mind when going up or down the stairs.

#### 1-3-1 Residual risk map



#### 1-3-2 Residual risk table

| No. | Operation step | Work                                    | Location in machine | Details of harm                                                                                                 | Protective measures to be taken by the customer                                                                                                                                                                                                                                                                                                 |
|-----|----------------|-----------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                |                                         | Seriousness of harm |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                 |
| 1   | Use            | Accessing the working or operation area | A                   | Tripping on or falling from the stairs                                                                          | <ul style="list-style-type: none"> <li>• Hold the handrail or the like when going up or down the stairs.</li> <li>• Do not run up or down the stairs.</li> </ul>                                                                                                                                                                                |
|     |                |                                         | Caution             |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                 |
| 2   | Installation   | Installing stairway by yourself         | A                   | If the stairs are not made or installed in accordance with ISO14122-3: 2016, their safety cannot be guaranteed. | <ul style="list-style-type: none"> <li>• If you prepare the stairs by yourself, make them in accordance with ISO14122-3: 2016.</li> <li>• When installing the stairs on the machine, be sure to check for safety.</li> </ul> <p><b>Remark :</b> When preparing the stairs by yourself, contact Mazak Technical Center or Technology Center.</p> |
|     |                |                                         | Notice              |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                 |

# 1 SAFETY PRECAUTIONS

## 1-4 Safety Nameplates

Because of its special characteristics, the machine requires extra care during operation. Safety nameplates for the prevention of occurrence of accidents, therefore, are attached to the dangerous sections of the body and peripheral units of the machine and to sections that may be used in the wrong way.

Always observe the precautions given on these safety nameplates and make it a practice of working safely.

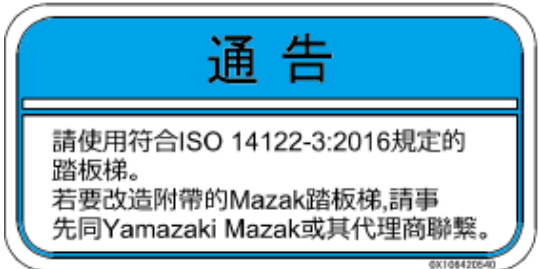
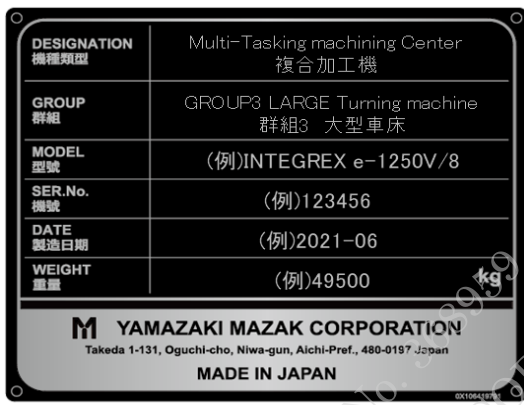
Typical safety nameplates are shown below. Understand that other safety nameplates are also the same in the degree of importance.

| Nameplate                                                                                            | Remarks                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a) White</p>   | <p>Falling from an elevated place (white)</p> <p>This plate is attached near the entrance to the stairs.</p> <p>Be careful not to fall from the stairs.</p> |
| <p>(b) Black</p>  | <p>Falling from an elevated place (black)</p> <p>This plate is attached near the entrance to the stairs.</p> <p>Be careful not to fall from the stairs.</p> |



## 1-5 Other Nameplates

Depending on the model, there is no explanation about the following nameplates in the Operating Manual, because they are not directly related to safety.

| Nameplate                                                                          | Remarks                                                                                                                                                      |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>The stairs are made in accordance with ISO14122-3: 2016. If there is a need to modify them, contact your Yamazaki Mazak dealer.</p>                       |
|  | <p>Printed on the nameplate are the destination, machine group name, model name, serial number, year and month of manufacture and weight of the machine.</p> |

## 1-6 Location of the Main Safety Devices

The machine has safety devices to ensure operator's and machine safety.  
(See Fig. 1-1 to Fig. 1-3.)

| No. | Name                          | Function                                                                                                                                                                                                                                                                                                                                                                            |
|-----|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Emergency stop button         | <ol style="list-style-type: none"> <li>Each manual operation panel is equipped with an emergency stop button.</li> <li>If the button is pressed, the machine will make an emergency stop regardless of whether it is operated manually or automatically.</li> </ol>                                                                                                                 |
| 2   | Operator door interlock       | <ol style="list-style-type: none"> <li>The operator door is locked in the closed position during automatic operation to protect the operator from a flying object.</li> <li>When the operator door is open and the machine setup switch is in the "I" position, the spindle and axes can be operated manually at a safe speed, but no other operations can be performed.</li> </ol> |
| 3   | Magazine door interlock       | The magazine does not operate when its door is open.                                                                                                                                                                                                                                                                                                                                |
| 4   | Pallet changer door interlock | The pallet changer does not operate when its door is open.                                                                                                                                                                                                                                                                                                                          |
| 5   | Machine setup switch          | Used to enable specific machine operations to be performed at a safe speed for setup with the operator door open.                                                                                                                                                                                                                                                                   |
| 6   | Enable button                 | This button is used along with another one to perform a specific operation. It is needed to perform a hold-to-run operation (operation that continues as long as the button is held down), such as operation for starting the spindle or moving an axis with the operator door open.                                                                                                |

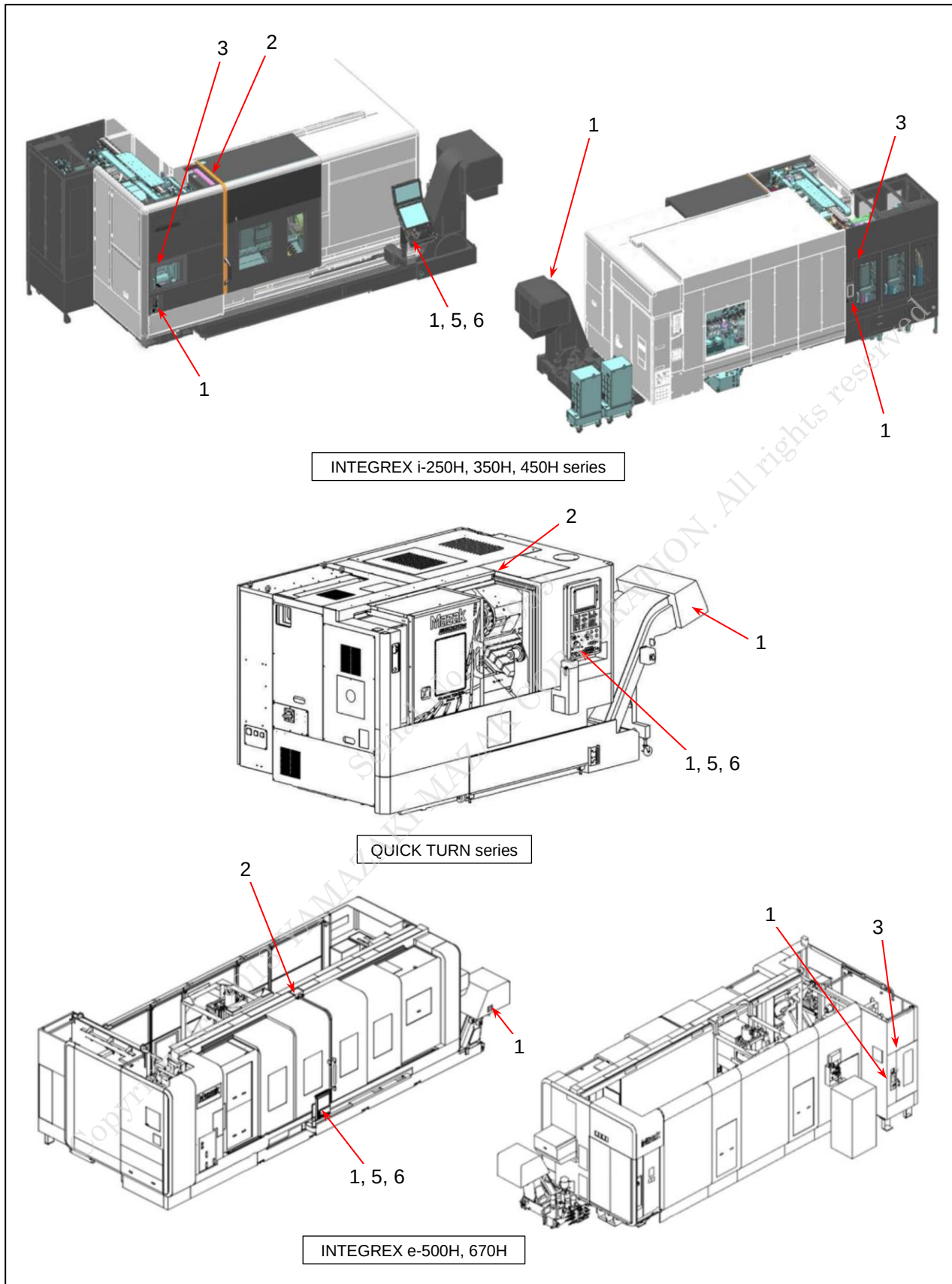


Fig. 1-1 Location of the safety devices (1/3)

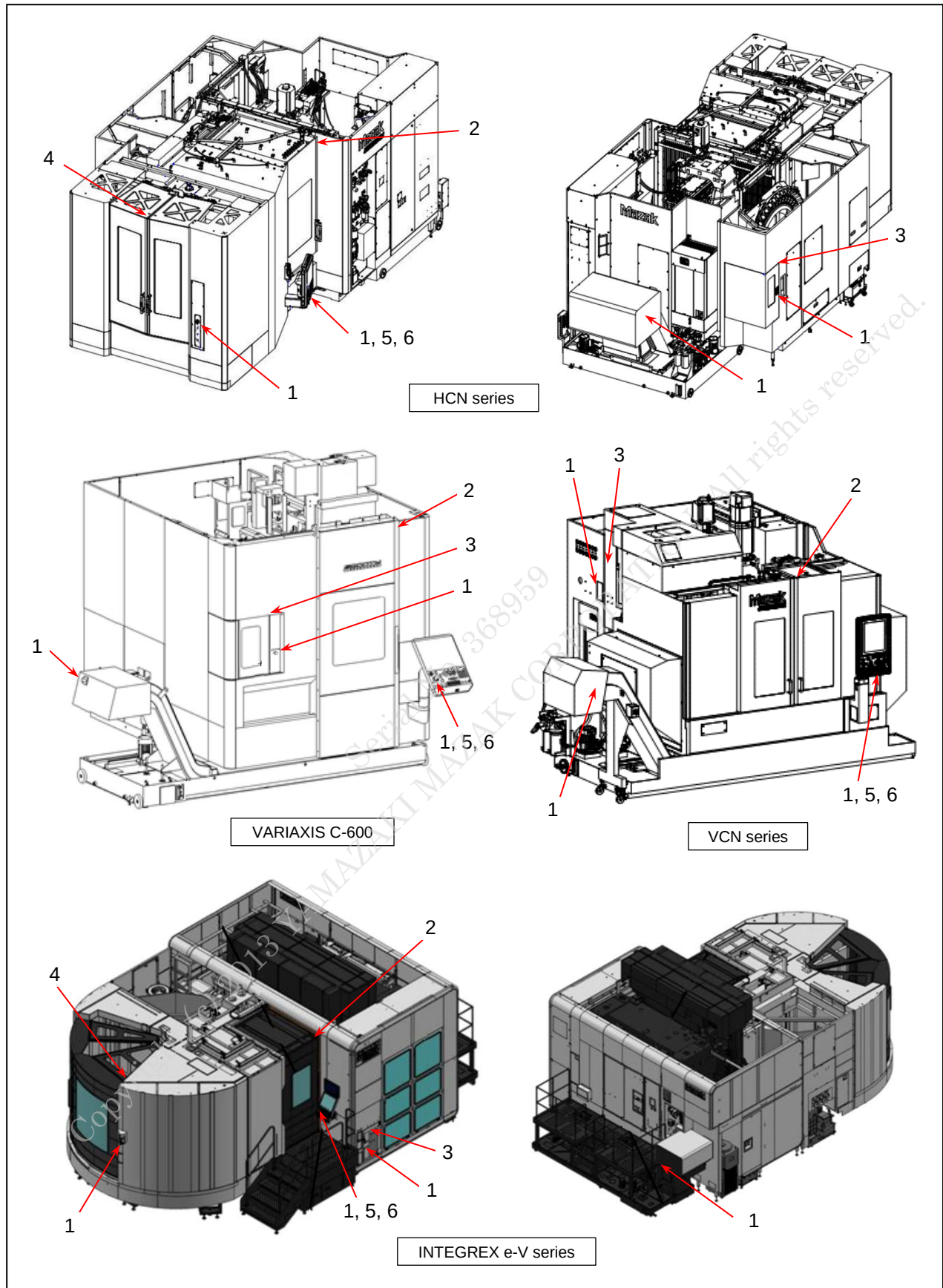


Fig. 1-2 Location of the safety devices (2/3)

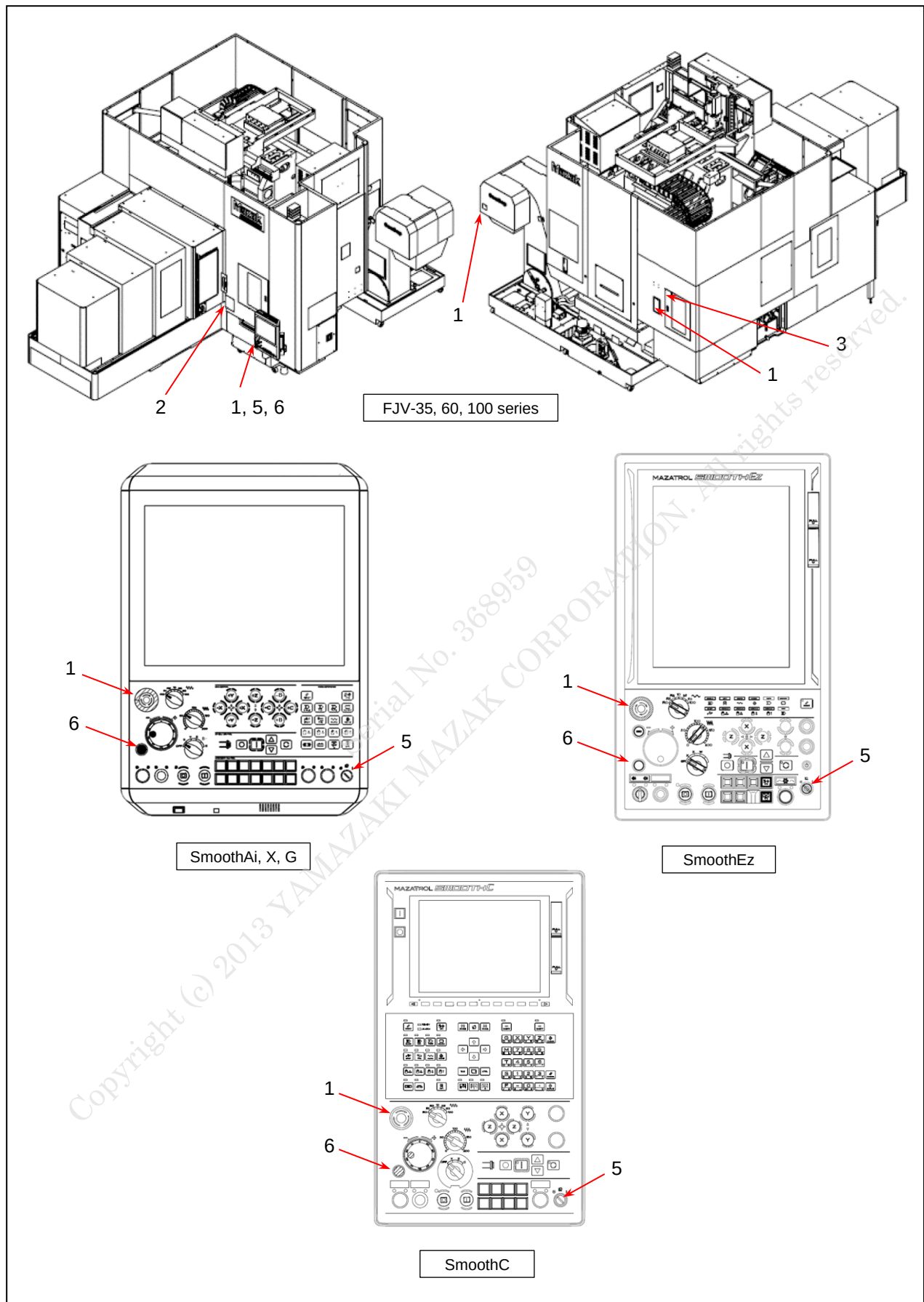


Fig. 1-3 Location of the safety devices (3/3)

1

SAFETY PRECAUTIONS

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## 2 MANUAL OPERATION WITH THE OPERATOR DOOR INTERLOCKED

### 2-1 Axis Movement/Spindle Rotation/TOOL EYE Operation

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>To perform the following manual operations when the operator door is open and the machine setup switch is in the "I" position, press the corresponding key while pressing the enable button. <ul style="list-style-type: none"> <li>Axis feed</li> <li>Spindle rotation</li> <li>TOOL EYE IN/OUT</li> </ul> </li> </ul> |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Note 1:** The enable button needs to be pressed simultaneously to operate the machine.

**Note 2:** If the enable button is not pressed as required, it will blink on and off.

|                                                                                   |                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>For an explanation of how the interlock works when the operator door is open, refer to the "Operator door interlock" section of the Operating Manual.</li> </ul> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2-2 Tailstock Operation With the Tail Spindle Switch

The targets of the operation are machines for which low-speed movement and high-speed movement can be switched in the tailstock operation with the tail spindle switch.

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>Regardless of the open/closed state of the operator door, to perform high-speed movement operation of the tailstock, operate the tail spindle switch while pressing the enable button instead of the  key. No workpiece can be pressed with this operation.</li> <li>When pressing a workpiece, operate the tail spindle switch without pressing the enable button and perform low-speed movement operation.</li> </ul> |
|  | <ul style="list-style-type: none"> <li>For the tailstock operation speed for each machine model, refer to the "Tailstock Operation" section of the Operating Manual.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     |





Blank area for manual operation with the operator door interlocked, featuring horizontal dashed lines for writing.

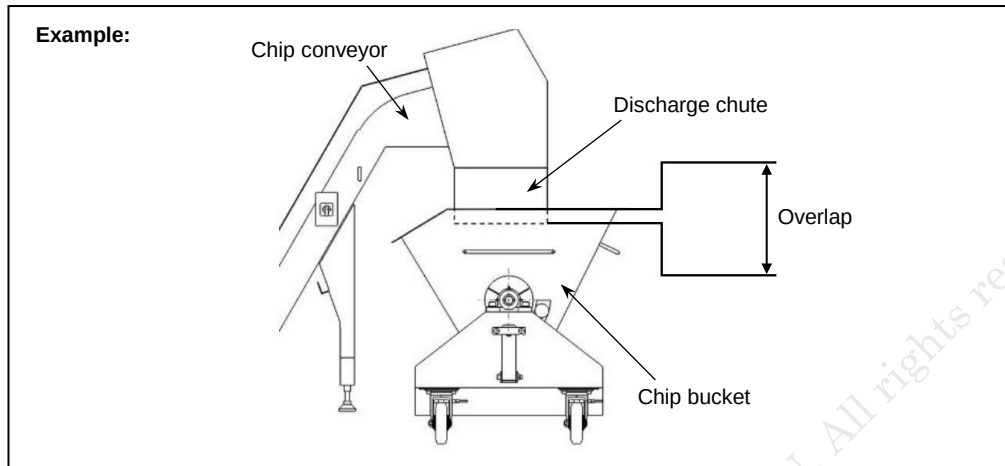
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### 3 NOTE ON THE CHIP BUCKET SETTING POSITION

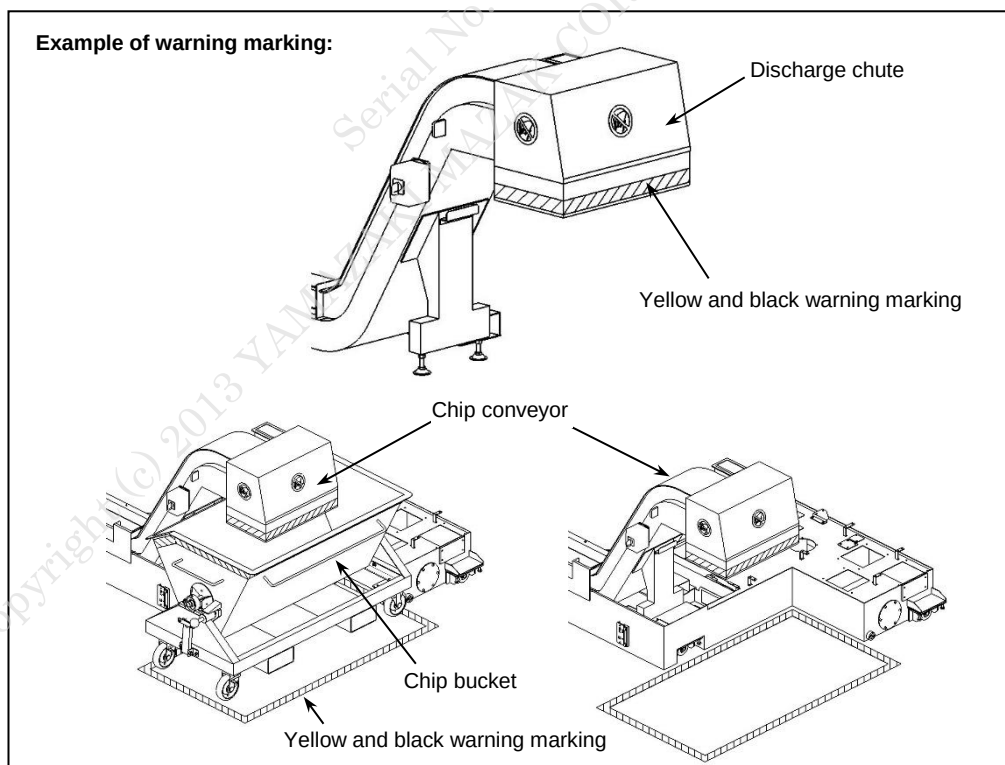


- If you put your hand in the discharge chute or the chip bucket, you may get seriously injured. Set the chip bucket so that at least 3 faces of it will overlap the discharge chute and the discharge chute will completely get in the chip bucket.



Mark the chip discharge area (around the discharge port or around the chip bucket setting position if a chip bucket is used) on the floor with a yellow and black warning line.

If the yellow and black warning line is faded or peeled off, mark the area again on your own responsibility.



**Remark :** Yellow and black warning marking tapes must be prepared by the customer. If they are difficult to obtain, contact Mazak Technical Center and Technology Center.

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NOTE ON THE CHIP BUCKET SETTING POSITION



Blank lined area for notes, consisting of multiple horizontal dashed lines.

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